

Dr. Vikram Goel

DESIGNATION	Scientist - D (Antarctic Cryospheric Studies Group)
CONTACT INFORMATION	National Centre for Polar & Ocean Research, +91-832-2525-707 Headland Sada, Vasco Da Gama, Goa 403804 vikram@ncpor.res.in vikram.goel@outlook.com ORCID: 0000-0002-8393-0195
RESEARCH INTERESTS	Antarctic Glaciology, Geophysical data collection and analysis (incl. ground-penetrating radar and GNSS systems), Ice-flow modelling, Geographical Information Systems (GIS).
EDUCATION	University of Oslo, Oslo, Norway Ph.D., Antarctic Glaciology, June 2019 • Thesis Topic: <i>Current status and past changes of ice rises in Dronning Maud Land, Antarctica</i> • Advisors: Kenichi Matsuoka, Jon Ove Hagen and Thamban Meloth Indian Institute of Technology Kharagpur, Kharagpur, WB, India M.Tech., Earth System Science & Technology, May 2013 • Thesis Topic: <i>Application of mid-latitude postfrontal shower characteristics to adverse weather avoidance model in aviation</i> • Advisors: Thomas Hauf, and M K Dash University of Petroleum and Energy Studies, Dehradun, UK, India B.Tech., Aerospace Engineering, May 2011 • Thesis Topic: <i>Analysis of advanced deep space propulsion techniques</i> • Advisors: Ugur Guven
RESEARCH EXPERIENCE	National Center for Polar & Ocean Research, Headland Sada, Vasco Da Gama, India Apr 2018 to Present Scientist - D (Nov 2023 - Present) Project Scientist - II (April 2022 - Nov 2023) Project Scientist - B (April 2018 to March 2022) • Led the collaborative (India-Norway) project SENS: Sub-shelf Exploration of Nivl Ice Shelf (as PI). During the successful SENS expedition on the Nivl Ice Shelf, the team conducted 130 km+ of seismic surveys using novel techniques and 500 km+ of radar surveys. • Led the geophysical survey component of the multinational (US-S.Korea-India-Norway-UK) collaborative field expedition RAICA-CP onboard the ice breaker RV/Araon. The team camped on Canisteo Peninsula ice rise in West Antarctica and drilled two 150-meter ice cores, complemented by 300 km+ of ground-based radar surveys. • Undertook the <i>41st Indian Scientific Expedition to Antarctica</i>. Conducted 500 km+ deep-sounding radar survey over Kamelryggen and Verbljud Ice Rises and other geophysical data. Leading the data analysis. • Undertook the <i>38th Indian Scientific Expedition to Antarctica</i> collecting various geophysical data in Nivlisen Ice Shelf and Leningrad Ice Rise. • Analyzed field-based and satellite data to determine mass balance of several ice rises.

- Investigated the past evolution of several ice rises by using an ice-flow model to simulate observed internal stratigraphy.
- Examined the regional characteristics of ice rises and suggested future direction of ice rise research.

Norwegian Polar Institute, Tromsø, Norway

Jan 2014 to Aug 2017

Research Fellow (Apr 2014 - Aug 2017)

- Investigated the current status and past evolution of ice rises in Dronning Maud Land, Antarctica.

Research Technician (Jan 2014 - Feb 2014)

- Participated in the *Norwegian Antarctic Research Expedition to Dronning Maud Land Antarctica*. Made glaciological measurements.

Leibniz University Hannover, Hannover, Germany

Sep 2012 to Mar 2013

Masters Student

- Studied the safety and efficiency relationship for an adverse weather avoidance model in aviation for mid-latitude postfrontal showers
Supervisor: Thomas Hauf

University of Massachusetts Dartmouth, MA, USA

Jun 2012 to Jul 2012

Masters Intern

- Worked on enhanced web availability of satellite derived SST data for western North Atlantic Ocean for local fishermen
Supervisor: James J. Bisagni

PUBLICATIONS

Peer-reviewed Papers

1. **Goel, V.**, Martín, C., Matsuoka, K., Pratap, B., Moholdt, G., Dey, R., Laluraj, C. M., and Thamban, M.: A new coastal ice-core site identified in Dronning Maud Land, Antarctica, for high-resolution climate reconstructions to the Last Glacial Maximum, *The Cryosphere*, 1–28, <https://doi.org/10.5194/egusphere-2025-2037>, 2025. (*accepted*)
2. Bingham, R. G., Bodart, J. A., Cavitte, M. G. P., Chung, A., Sanderson, R. J., Sutter, J. C. R., Eisen, O., Karlsson, N. B., MacGregor, J. A., Ross, N., Young, D. A., Ashmore, D. W., Born, A., Chu, W., Cui, X., Drews, R., Franke, S., **Goel, V.**, Goodge, J. W., Henry, A. C. J., Hermant, A., Hills, B. H., Holschuh, N., Koutnik, M. R., Leysinger Vieli, G. J.-M. C., MacKie, E. J., Mantelli, E., Martín, C., Ng, F. S. L., Oraschewski, F. M., Napoleoni, F., Parrenin, F., Popov, S. V., Rieckh, T., Schlegel, R., Schroeder, D. M., Siegert, M. J., Tang, X., Teisberg, T. O., Winter, K., Yan, S., Davis, H., Dow, C. F., Fudge, T. J., Jordan, T. A., Kulesa, B., Matsuoka, K., Nyqvist, C. J., Rahnmooonfar, M., Siegfried, M. R., Singh, S., Višnjević, V., Zamora, R., and Zuhr, A.: Review article: AntArchitecture – building an age–depth model from Antarctica’s radiostratigraphy to explore ice-sheet evolution, *The Cryosphere*, 19, 4611–4655, <https://doi.org/10.5194/tc-19-4611-2025>, 2025.
3. Pritchard, H. D., Fretwell, P. T., Fremand, A. C., Bodart, J. A., Kirkham, J. D., Aitken, A., Bamber, J., Bell, R., Bianchi, C., Bingham, R. G., Blankenship, D. D., Casassa, G., Christianson, K., Conway, H., Corr, Hugh F. J., Cui, X., Damaske, D., Damm, V., Dorschel, B., Drews, R., Eagles, G., Eisen, O., Eisermann, H., Ferraccioli, F., Field, E., Forsberg, R., Franke, S., **Goel, V.**, and 53 others: Bedmap3 updated ice bed, surface and thickness gridded datasets for Antarctica, *Scientific Data*, 12, 414, <https://doi.org/10.1038/s41597-025-04672-y>, 2025.

4. **Goel, V.**, Martín, C., and Matsuoka, K.: Evolution of ice rises in the Fimbul Ice Shelf, Dronning Maud Land, over the last millennium, *Antarctic Science*, 1–15, <https://doi.org/10.1017/S0954102023000330>, 2024.
5. Cavitte, M. G. P., Goosse, H., Matsuoka, K., Wauthy, S., **Goel, V.**, Dey, R., Pratap, B., Van Liefferinge, B., Meloth, T., and Tison, J.-L.: Investigating the spatial representativeness of Antarctic ice cores: A comparison of ice core and radar-derived surface mass balance, *The Cryosphere*, <https://doi.org/10.5194/tc-17-4779-2023>, 2023.
6. Frémand, A. C., Fretwell, P., Bodart, J., Pritchard, H. D., Aitken, A., Bamber, J. L., Bell, R., Bianchi, C., Bingham, R. G., Blankenship, D. D., Casassa, G., Catania, G., Christianson, K., Conway, H., Corr, H. F. J., Cui, X., Damaske, D., Damm, V., Drews, R., Eagles, G., Eisen, O., Eisermann, H., Ferraccioli, F., Field, E., Forsberg, R., Franke, S., Fujita, S., Gim, Y., **Goel, V.**, Gogineni, S. P., and 50 others: Antarctic Bedmap data: FAIR sharing of 60 years of ice bed, surface and thickness data, *ESSD – Ice/Glaciology*, <https://doi.org/10.5194/essd-15-2695-2023>, 2023.
7. **Goel, V.**, Morris, A., Moholdt, G., and Matsuoka, K.: Synthesis of field and satellite data to elucidate recent mass balance of five ice rises in Dronning Maud Land, Antarctica, *Frontiers in Earth Science*, 10, <https://doi.org/10.3389/feart.2022.975606>, 2022.
8. Pratap, B., Dey, R., Matsuoka, K., Moholdt, G., Lindback, K., **Goel, V.**, Laluraj, C. M., and Meloth, T.: Three-decade spatial patterns in surface mass balance of the Nivlisen Ice Shelf, central Dronning Maud Land, East Antarctica, *Journal of Glaciology*, 1–13, <https://doi.org/10.1017/jog.2021.93>, 2021.
9. **Goel, V.**, Matsuoka, K., Berger, C. D., Lee, I., Dall, J., and Forsberg, R.: Characteristics of ice rises and ice rumpled in Dronning Maud Land and Enderby Land, Antarctica, *Journal of Glaciology*, 1–15, <https://doi.org/10.1017/jog.2020.77>, 2020.
10. Morlighem, M., Rignot, E., Binder, T., Blankenship, D., Drews, R., Eagles, G., Eisen, O., Ferraccioli, F., Forsberg, R., Fretwell, P., **Goel, V.**, Greenbaum, J. S., and 25 others: Deep glacial troughs and stabilizing ridges unveiled beneath the margins of the Antarctic ice sheet, *Nature Geoscience*, 1–6, <https://doi.org/10.1038/s41561-019-0510-8>, 2019.
11. **Goel, V.**, Martín, C., and Matsuoka, K.: Ice-rise stratigraphy reveals changes in surface mass balance over the last millennia in Dronning Maud Land, *Journal of Glaciology*, 64(248), 932–942, <https://doi.org/10.1017/jog.2018.81>, 2018.
12. **Goel, V.**, Brown, J., and Matsuoka, K.: Glaciological settings and recent mass balance of Blåskimen Island in Dronning Maud Land, Antarctica, *The Cryosphere*, 11, 2883–2896, <https://doi.org/10.5194/tc-11-2883-2017>, 2017.

Published Data

Published datasets at NPI data center (data.npolar.no)

White Papers

1. Matsuoka, K., Clifton, R., Blankenship, D., Cui, X., Eagles, G., Eisen, O., **Goel, V.** and 15 others: International collaboration is urgently needed for airborne surveys in Enderby Land to fill the largest RINGS data gap. *Zenodo*. <https://doi.org/10.5281/zenodo.7818281> , 2023.

AWARDS

- Awarded *Outstanding Employee Award 2025* at NCPOR
- Awarded travel support of USD850 from the SCAR RINGS action group to attend the SCAR 2024 meeting and associated workshops
- Travel award of 7000 Norwegian Kroner (700 Euro) to attend SCAR (Scientific Committee on Antarctic Research) RINGS Action Group Workshop in Tromsø, Norway, June 2022
- Travel award of \$500 from the SCAR Physical Sciences group to attend Polar 2018, Davos, Switzerland, Jun 2018.
- Best oral presentation at the International Glaciological Society Nordic Branch Meeting, Reykjavik, Iceland, Oct 2014
- DAAD (German Academic Exchange Service) IIT Master Thesis Scholarship

CONFERENCES

Talks

- SaGAA VIII Leadership Conference
Title: *"Accelerating Coastal Antarctic Research: Learning from Recent Field Innovations"* Nov 2025
- Workshop for DML/EL RINGS, Institute of Low Temperature Science, Hokkaido University, Japan
Title: *"SENS: Sub-shelf Exploration of Nivl Ice Shelf"* Sep–Oct 2025
Title: *"Surface Clutter Modeling in Radar Sounding"*
- NCPS 2025, NCPOR, Goa, India
Title: *"Sub-Shelf Exploration of Nivlisen Ice Shelf, East Antarctica: Initial results from geophysical surveys"* Sep 2025
- SCAR Open Science Conference 2024, Pucon, Chile
Title: *"Coastal ice core site identified in Dronning Maud Land, Antarctica"* Aug 2024
- Invited talk at 2nd RINGS Workshop
Title: *"Constraining regional SMB using Radar and Ice Cores"*
- SCAR Open Science Conference 2022 Online
Title: *"Glaciological site survey to determine an optimal coring site in coastal Dronning Maud Land, Antarctica "* Aug 2022
- IGS Nordic Branch Meeting 2020 Online
Title: *"Characteristics of ice rises in Dronning Maud Land, Antarctica"* Nov 2020
- SCAR Open Science Conference 2020 Online
Title: *"Present and past evolution of Fimbul Ice Shelf region, East Antarctica: Ground-based investigations over three ice rises"* Aug 2020
- Climate Change Impacts, Vulnerabilities, and Adaptation (CCIVA), Indian Institute of Technology, Kharagpur, West Bengal, India
Title: *"Characteristics of ice rises and rumples in Dronning Maud Land, Antarctica"* Mar 2019
- International Glaciological Society Nordic Branch Meeting, Tromsø, Norway
Title: *"Glaciological settings and recent mass balance of Blaskimen Ice Rise, Antarctica"* Oct 2016
- International Glaciological Society International Symposium on Interactions of Ice Sheets and Glaciers with the Ocean, San Diego, USA
Title: *"Glaciological settings and recent mass balance of Blaskimen Ice Rise, Antarctica"* Jul 2016
- 12th International Symposium on Antarctic Earth Sciences, Goa, India
Title: *"Recent thickening of the Blaskimen Ice Rise in Dronning Maud Land Antarctica"* Jul 2015
- International Glaciological Society Nordic Branch Meeting, Reykjavik, Iceland
Title: *"Current status and past changes of ice rises in Dronning Maud Land, Antarctica– Initial results from Blaskimen"* Oct 2014

Posters

- SCAR Open Science Conference 2024, Pucon, Chile
Title: *"Retrieving High-Resolution Ice Core Records and Radar Stratigraphy from*

	<i>Canisteo Peninsula, West Antarctica</i>	Aug 2024
	• SCAR RINGS Action Group Workshop Title: <i>“Geophysical survey to determine an optimal ice coring site in coastal Dronning Maud Land”</i>, at the SCAR RINGS Workshop, Tromsø, Norway June 2022 • <i>“Current Status and Past Evolution of Ice Rises in Dronning Maud Land, Antarctica”</i>, at the Polar 2018 conference, Davos, Switzerland Jun 2018 • <i>“This is APECS (Association of Polar Early Career Scientists)”</i>, at the CHES all staff meeting, Bergen, Norway Mar 2016	
LEADERSHIP AND SERVICE	• National Representative to the SCAR GeoSciences Group , 2024 - Present Aug • NCPS 2025, NCPOR, Goa Sep 2025 • Led the poster session organisation team • ATCM 2024: Kochi 2024 May 2024 • Head coordinator (Outreach and media) for the local organising committee • First point of contact to the ATCM chair • Early career researcher representative to the SCAR’s RINGS Action Group, Oct 2021 - Present • Co-organized and chaired a panel discussion on ‘Lessons from previous projects’ during the SCAR RINGS Action Group workshop, Tromsø, Norway, June 2022 • Led organising and chairing of Session 10 during the SCAR Open Science Conference 2020 Online, Aug 2020 • Part of APECS (Association of Polar Early Career Scientists) leadership as a council member from 2015-2021 • Led the project group to organise <i>“APECS annual online conference 2019: Breaking Barriers”</i> • Co-organized and chaired a breakout session during the APECS World Summit 2018, Davos, June 2018 • Organized <i>APECS Panel discussion on “Work-Personal life balance”</i> during the IGS Nordic Branch Meeting, Tromsø, 2016	
	Sessions chaired	
	• <i>Chaired the session 12 during SCAR Open Science Conference 2024.</i> Aug 2024 • <i>Chaired a session during SCAR Open Science Conference 2020 Online</i> Aug 2020 • <i>Chaired a session during APECS annual online conference 2019: Breaking Barriers</i> Mar 2018 • <i>Chaired a session during APECS annual online conference 2018: Butterfly effect</i> Mar 2018 • <i>Chaired a session during APECS annual online conference 2017: Outside the box</i> Mar 2017 • <i>Chaired a session at the International Glaciological Society Nordic Branch Meeting, Tromsø, Norway</i> Oct 2016 • <i>Chaired a session at the CHES all staff meeting, Bergen</i> Mar 2016	
	Journal Reviewer	
	• Journal of Glaciology • Earth and Space Science • Journal of Geophysical Research: Earth Surface • The Cryosphere	

- NEWS AND MEDIA
- *The Print: NCPOR in Goa studies extremes from Antarctica to deep ocean research* Jan 2026
 - *Science: Daring James Bond mission to drill Antarctic ice cores could reveal future sea level rise* Apr 2024
 - *Indian Express: In Antarctica ice, hunt for climate change clues* Jan 2018
 - *Scroll: An Indian scientist is studying the South Pole from the North Pole* Feb 2017